

Karan Bhukar

MSCS STUDENT, NYU COURANT

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Education

New York University Courant

MASTER OF SCIENCE IN COMPUTER SCIENCE

CGPA: NA/4.00

September 2026 - Present

Indian Institute of Technology Hyderabad

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE & ENGG.

CGPA: 9.04/10.00

Aug 2018 - June 2022

Work Experience

Riviere Robot Lab, Center for Robotics and Embodied Intelligence, NYU

GRADUATE RESEARCHER MENTORED BY PROF. BENJAMIN RIVIERE

New York

June 2026 - Present

• Quantifying Inference-Time Planning Gains via Offline Dataset Coverage [Ongoing Research]

- Formulating a dataset coverage metric ($G(D)$) to quantify when online planning resolves offline RL pessimism and distributional shift.
- Leveraging a PyTorch GPU-accelerated MPPI planner warm-started with IQL value priors as an experimental testbed for coverage expansion.
- Designing synthetic out-of-distribution (OOD) dataset gaps to isolate and measure local coverage expansion (ΔCov).
- Evaluating the correlation between local dataset coverage gains and planner value advantage ($V_{\pi_\mu} - V_{\pi_\theta}$) to predict MPC compute utility.

Amazon

APPLIED SCIENTIST

Bengaluru, India

Oct 2024 - Aug 2026

• Packed Masked Diffusion for High-Throughput Search Query Understanding

(AMLC 2026) [C.6]

- Architected a novel packed **masked diffusion LM framework** to accelerate search query understanding at scale.
- Achieved 9.8 times higher throughput** wrt to autoregressive models while maintaining standard baseline accuracy.
- Implemented a custom block diagonal attention mask to strictly enforce query isolation across concatenated batches
- Generated $\sim 5\text{B}$ training sequences spanning 100M+ entities by building an LLM and metadata-driven data pipeline.
- Boosted **core search metric (SRPR)** by **151 bps** in South America and **61 bps** in North America, directly improving user search discovery.

• Popularity Bias in Music Search Ranker (with Prof. Arpit Agarwal, IIT Bombay)

(Under Submission AAAI 2027) [M.7]

- Directed the **Amazon-IIT Bombay** research collaboration, leading a multi-institutional team to investigate bias in Recommender systems.
- Co-developed a continuous-time ODE framework to model user-recommender co-evolution dynamics.
- Co-Derived critical imbalance thresholds (p) and provable guarantees for symmetric user retention.
- Validated on 410M+ Amazon Music interactions, quantifying niche user churn and demonstrating equalized-odds mitigation strategies.

IBM Research

RESEARCH ENGINEER MENTORED BY HARSHIT KUMAR, DINESH RAGHU

Gurgaon, India

July 2021 - Oct 2024

• BERTOps, Pretrain-to-Production Log Analytics (Demo:)

(Oral at IEEE CLOUD '23, ICPE '25 & IAAI '26) [C.2] [C.4] [C.5] [P.6]

- Built BERTOps, a domain-specialized masked-LM for semi-structured IT logs, to capture template regularities that general LLMs miss.
- Released labeled datasets and a unified benchmark for Log Format Detection, Golden Signal Classification, and Fault Category Prediction including the first public labels for GSC/FCP.
- Productionized models into a log-analytics tool; optimized for CPU-only model inference for millions of logs via log templating.
- At production scale, saved $\sim 8\text{K}$ engineer-minutes by processing 1.04B log lines (877 GB) across 100+ internal software support teams.

• False Alert Suppression on Observability Platform (Demo:)

(Oral at ICSE '24 and Workshop at COMSNET '24) [C.3] [W.1] [P.3] [P.5]

- Designed *Dynamic-X-Y*: an unsupervised alert-suppression policy learner using moving-average envelopes over historical alert streams; applies per-signal dynamic thresholds at runtime.
- On log and metric anomaly datasets, improved accuracy over baselines by **+7.39%** and **+35.7%**, respectively.
- Built a train-to-deploy pipeline that learns per-signal X-in-Y suppression thresholds from historical alerts and pushes them as runtime policies via AI Model Management.
- In internal deployments, cut non-actionable incident notifications by $\sim 90\%$ while preserving critical alerting.

• Reinforcement Learning for Conversation Disentanglement(CD)

(Oral & Poster at AAAI '23; Patents in US, CN & JP) [C.1] [P.1] [P.2]

- Framed CD as end-to-end RL problem, optimizing a global conversation-level objective over local link predictions.
- Designed a global clustering-based reward and a policy that sequentially assigns messages to threads.
- Proposed a novel hybrid reward balancing the global disentanglement metric with local link-level accuracy.
- Translated the research into applied systems, resulting in two patent applications one pursued internationally (US/CN/JP) and one US-only.
- This research was partly conducted during my internship.

Publications

C=Conference Paper, W=Workshop Paper, M=Manuscript. * denotes Equal Contribution

[M.7] Stay or Stray-A Dynamical Systems Viewpoint of Popularity Bias 

Shashidhar S., Prabhat L., Agarwal A., Manjunath D., Bhukar K., Khandelwal T.

(arXiv preprint 2026)

[C.6] Packed Masked Diffusion for High Throughput Query Understanding in Music Search 

Bhukar K.*, Tandon P.*, Garg A., Khandelwal T.

(AMLC 2026)

[C.5] Scalable and Efficient Large-Scale Log Analysis with LLMs: An IT Software Support Case Study  Gupta P., **Bhukar K.**, Kumar H., Nagar S., Mohapatra P., Kar D. (40th AAAI 2026)

[C.4] LogAn: An LLM-Based Log Analytics Tool with Causal Inferencing  Gupta P., **Bhukar K.**, Kumar H., Nagar S., Mohapatra P., Kar D. (25th ICPE 2025)

[C.3] Dynamic Alert Suppression Policy for Noise Reduction in AIOps  **Bhukar K.**, Kumar H., Mahindru R., Nagar S., Aggarwal P., Arora R., Paradkar A. (46th ICSE 2024)

[W.1] Dynamic-X-Y: A Tool for Learning Dynamic Alert Suppression Policies in AIOps  **Bhukar K.**, Kumar H., Arora R., Mahindru R., Nagar S., Thornhill M., Manning I., Buggins J., Cook S., Paradkar A., Aggarwal P. (16th COMSNETS 2024)

[C.2] Learning Representations on Logs for AIOps  Gupta P., Kumar H., Kar D., **Bhukar K.**, Aggarwal P., Mohapatra P. (16th IEEE Cloud 2023)

[C.1] End-to-End Deep Reinforcement Learning for Conversation Disentanglement  **Bhukar K.**, Kumar H., Raghu D., Gupta A. (37th AAAI 2023)

[M.1] Target-Agnostic Adversarial Attacks on Language Understanding Models  Chauhan J., **Bhukar K.**, Kaul M. (arXiv preprint 2021)

Patents

[P.6] S&M for Logline Enrichment for Error Analysis P=Patent, S&M=Systems and Methods **Bhukar K.**, Kumar H., Nagar S., Gupta P., Samanta S., Mohapatra p., Kar D., Aggarwal P. U.S. Patent under filing 2025

[P.5] S&M for Explainability-Driven Control of an Alert Suppression Policy  **Bhukar K.**, Kumar H., Nagar S., Mahindru R., Paradkar A., Manning I., Hussey S. U.S. Patent 2026

[P.4] S&M for Determining Log Templates of a Computing Device for Log Templatization  **Bhukar K.**, Gupta P., Kumar H., Verma M. U.S. Patent under filing 2024

[P.3] S&M for Dynamic Policy Management for Events and Alerts Noise Reduction in IT systems  **Bhukar K.**, Nagar S., Kumar H., Aggarwal P., Mahindru R., Paradkar A., Manning I., Thornhill M., Arora R., Hussey S., Forti F. U.S. Patent 2023

[P.2] S&M To Recommend/Revoke Agents Guided by the AIOps Conversation Context  **Bhukar K.**, Kumar H., Azad A., Joshi S. U.S. Patent 2023 & CN/JP Patent 2024

[P.1] S&M to Automatically Generate Close Notes from SRE Interactions for Downstream AIOps Tasks  **Bhukar K.**, Kumar H., Azad A., Joshi S. U.S. Patent 2023

Skills and Frameworks

Languages Proficient in: Python, C++; Familiar with: SQL, Java, JavaScript, Haskell, Prolog, HTML, CSS, Eiffel, MATLAB

Software & Tools JAX, PyTorch, TensorFlow, Transformers, Pyspark, Spark, Django, AWS, $\LaTeX$, Git, Vim

Talks & Presentations

- 2024 46th International Conference on Software Engineering, @ Centro Cultural de Belém, Lisbon, Portugal
- 2024 IBM Research Industry Expo Booth, @ 37th NeurIPS, New Orleans, USA
- 2023 37th AAAI Conference on Artificial Intelligence, @ Walter E. Washington Convention Center, Washington, D.C., USA

Scholastic Achievements

- 2024 Outstanding Technical Achievement Award (OTAA), IBM Research’s highest distinction for technical excellence, recognizing innovative work that enabled multi-million-dollar business growth.
- 2024 Entrepreneurship Award, awarded at IBM for end-to-end ownership of a project that matured into a major internal product.
- ’22-24 6 Inventor Award, acknowledged at IBM for successful filing of multiple patents in USPTO
- 2021 Among 50, from 10,000+ applicants selected for the prestigious IBM Research Blue Scholar Internship Program.
- 2018 All India Rank of 681, among 0.15 million candidates in the JEE Advanced conducted by the Joint Admission Board.
- 2018 All India Rank of 358, among 1.5 million candidates in the JEE Main, conducted by CBSE.
- 2018 KVPY Fellowship, by the Dept. of Science, Government of India, among top 2000 out of 100,000+ applicants.
- 2016 NTSE Scholarship, Ranked 5, in Rajasthan state conducted by NCERT, Government of India, among 100,000+ applicants.

Volunteering & Community Service

- 2024 Reviewer @ AAAI ’24, for submissions pertaining to AAAI ’23 research track
- 2024 Internship Mentor @ IBM Research, for Shivangi Shreya, CS Masters student at IIT Madras
- 2020-21 Volunteer @ COLING ’20 & NeurIPS ’21, assisted in the online conference organization during covid
- Fall ’21 Teaching Assistant @ Dept. of CSE, IIT-H, Foundations of ML, lectured by Prof. Vineeth N. Balasubramanian
- Fall ’20 Teaching Assistant @ Dept. of CSE, IIT-H, Data Structures, lectured by Prof. Karteek Sreenivasaiah
- Spring ’19 Teaching Assistant @ Dept. of CSE, IIT-H, Artificial Intelligence, lectured by Prof. P. K. Srijith